8. Responsibilities of a Competing Consolidator

The Commission proposed Rule 614(d) to establish the responsibilities applicable to all competing consolidators, including competing consolidators that are affiliated with SROs and those that are not, under the decentralized consolidation model. Under proposed Rule 614(d), all competing consolidators would be required to perform many of the obligations currently performed by the existing exclusive SIPs. Proposed Rule 614(d) also would require all competing consolidators to disclose performance metrics and other information that would facilitate Commission oversight of competing consolidators and assist market participants in evaluating and choosing competing consolidators.

(a) Rules 614(d)(1) through (3): Collection, Calculation, and Dissemination of Consolidated Market Data

(i) Proposal

Proposed Rules 614(d)(1) through (3) would require competing consolidators to collect, consolidate, and disseminate consolidated market data. Proposed Rule 614(d)(1) would require each competing consolidator to collect from each national securities exchange and national securities association, either directly or indirectly, the information with respect to quotations for and transactions in NMS stocks as provided in Rule 603(b). Proposed Rule 614(d)(2) would require each competing consolidator to calculate and generate consolidated market data, as defined in proposed Rule 600(b)(19), from the information collected in proposed Rule 614(d)(1). This proposed rule would require competing consolidators to develop a complete consolidated market data product that contained all of the data elements specified in the proposed definition of consolidated market data.[1003] Proposed Rule 614(d)(3) would require competing consolidators to

[1003] See Proposing Release, 85 FR at 16782.

make the proposed consolidated market data available to subscribers on a consolidated basis and on terms that are not unreasonably discriminatory, with the timestamps required by proposed Rule 614(d)(4) and Rule 614(e)(1)(ii), as discussed below.

(ii) Final Rule and Response to Comments

The Commission received several comments regarding proposed Rules 614(d)(1) through (3).[1004] One commenter strongly supported requiring competing consolidators to be subject to appropriate standards, such as providing fair access to market participants.[1005] Two commenters criticized the proposed rules. One commenter stated that incorporating aggregated odd-lot quotes into the NBBO calculation would cause confusion and suggested instead that the NBBO be based on exchange BBOs "to minimize any calculation or interference/influences" by competing consolidators.[1006] With respect to proposed Rule 614(d)(2), one commenter said that because the Commission did not define what it meant by "generate" proposed consolidated market data, it was unclear what types of activity would warrant registration by competing consolidators.[1007] The commenter also argued that the Commission did not describe the meaning of "unreasonably discriminatory" in proposed Rule 614(d)(3),[1008] the consequences for

---

1004 See Clearpool Letter at 7; Data Boiler Letter I at 52; IDS Letter I at 10–11.

1005 See Clearpool Letter at 7.

1006 Data Boiler Letter I at 52. This commenter also suggested that only a single competing consolidator should be obligated to provide a consolidated market data product, not all competing consolidators. See supra Section III.C.1(b).

1007 IDS Letter I at 11.

1008 The commenter said that data vendors are currently not held to an "unreasonably discriminatory" standard. Because the commenter believed the Commission did not explain how it would apply this standard, the commenter said that commenters cannot

competing consolidators that make data available on an unreasonably discriminatory basis, the "costs of the mechanisms and consequences for application and enforcement of the unreasonably discriminatory requirement for both the relevant competing consolidator and its clients," and whether agreements between a competing consolidator and its subscribers that limit the competing consolidator's liability would be deemed "unreasonably discriminatory."[1009]

As discussed above,[1010] the Commission is not requiring competing consolidators to sell a full consolidated market data product and is amending proposed Rules 614(d)(1) through (d)(3) to reflect this change. Rule 614(d)(1), as amended, requires each competing consolidator to collect from each national securities exchange and national securities association, either directly, or indirectly, only the information required under Rule 603(b) that is necessary for the competing consolidator to create the particular consolidated market data product(s) it chooses to sell. Rule 614(d)(2), as amended, requires each competing consolidator to calculate and generate a consolidated market data product from the data collected pursuant to Rule 614(d)(1). Rule 614(d)(3), as amended, requires each competing consolidator to make the consolidated market data product(s) available to subscribers on terms that are not unreasonably discriminatory, and timestamped as required by Rule 614(d)(4) and including the national securities exchange and national securities association generated timestamp as required by Rule 614(e)(1)(ii).

With respect to the comment that the NBBO should be based on exchange BBOs and that incorporation of aggregated odd-lot quotes would cause confusion,[1011] the Commission believes

---

assess the standard's burden on potential competing consolidators, including data vendors. Id. at 10.

1009 Id. at 10–11.

1010 See supra Section II.B.2; see also supra Section III.C.1(b). The Commission is adopting a definition for "consolidated market data product" in Rule 600(b)(20).

1011 See Data Boiler Letter I at 52.

that requiring the exchanges to calculate their BBOs before sending them to the competing consolidators for calculation into NBBOs would add latency to the collection, consolidation, and dissemination of consolidated market data products. One of the goals of the introduction of competing consolidators is the reduction of latencies. Any exchange processing of the data content underlying consolidated market data will add latency in the collection, calculation, and dissemination of consolidated market data. Further, the exchanges currently aggregate odd-lot quotes into their BBOs, which are used to calculate the NBBO. Therefore, the Commission does not believe that this provision will cause confusion.[1012] The Commission believes that aggregating odd-lots into the BBO provides market participants with a more complete view of the market for each security.

In response to the comment that only one competing consolidator should be obligated to provide a consolidated market data product,[1013] the Commission is not requiring all competing consolidators to provide all consolidated market data. Competing consolidators may sell a consolidated market data product comprising some or all components of consolidated market data.[1014] The Commission believes that competing consolidators should have the flexibility to tailor their market data products to their subscribers' needs. Not requiring competing consolidators to sell a product that contains all of the data elements of full consolidated market data should enhance competition among competing consolidators by providing more parameters (e.g., products) upon which they can compete.

1012 See supra Section III.B.10.

1013 See Data Boiler Letter I at 52.

1014 See Rule 600(b)(20).

With respect to the comment that requested clarification of the use of the term "generate" in proposed Rule 614(d)(2), competing consolidators will be required to calculate and generate a consolidated market data product from the individual data streams made available by the SROs pursuant to Rule 603(b). For example, competing consolidators that choose to sell a consolidated market data product that includes the NBBO will calculate the NBBO as set forth in Rule 600(b)(50) and disseminate the NBBO in the consolidated market data product. Competing consolidators that sell a consolidated market data product that includes depth-of-book data will generate depth-of-book data by considering the NBBO and then determining the five price levels above (below) the NBBO from the quotation information provided by the SROs. The "calculate and generate" description refers to the processes that competing consolidators will use to create a consolidated market data product from the individual SRO quotation and transaction information they receive. Competing consolidators that receive transaction and quotation information from the individual SROs pursuant to Rule 603(b) and calculate and generate a consolidated market data product for dissemination must register pursuant to Rule 614.

"Unreasonably discriminatory" is a term used in Section 11A(c)(1)(D) of the Exchange Act.[1015] Section 11A(c)(1)(D) of the Exchange Act states that all exchange members, brokers, dealers, SIPs, and, subject to such limitations that the Commission may impose as necessary or appropriate for the protection of investors or maintenance of fair and orderly markets, all other persons may obtain on terms that are not unreasonably discriminatory such information with respect to quotations for and transactions in any security, other than an exempted security, as is

[1015] 15 U.S.C. 78k-1(c)(1)(D). See also Bloomberg Order, supra note 22.

published or distributed by any SRO or SIP. The term "unreasonably discriminatory" in Rule 614(d)(3) has the same meaning as in Section 11A(c)(1)(D). With respect to the comment asking about the applicability of this term,[1016] while such determinations are facts and circumstances-based and specific to each individual situation, a competing consolidator should have a reasonable basis for providing a consolidated market data product on different terms to different customers.

(b) Rule 614(d)(4): Timestamping of Consolidated Market Data

(i) Proposal

Proposed Rule 614(d)(4) would require each competing consolidator to timestamp the information collected in proposed Rule 614(d)(1): (i) upon receipt from each national securities exchange and national securities association at the exchange's or association's data center; (ii) upon receipt of such information at its aggregation mechanism; and (iii) upon dissemination of consolidated market data to customers.

(ii) Final Rule and Response to Comments

The Commission received several comments regarding proposed Rule 614(d)(4).[1017] Three commenters supported the timestamp requirement of the proposed Rule.[1018] One commenter said that market participants could use the originating venue timestamp and the consolidator timestamps to gauge whether the latency meets their needs and whether their best-

---

1016 See IDS Letter I at 10–11.

1017 See Capital Group Letter at 4; IEX Letter at 8; Data Boiler Letter I at 53; TD Ameritrade Letter at 13.

1018 See Capital Group Letter at 4; IEX Letter at 8; Data Boiler Letter I at 53.

execution obligations were met.[1019] Another commenter suggested a time granularity of +/- 50 milliseconds or in sub-milliseconds.[1020]

One commenter, however, stated that the proposed rule could create confusion.[1021] This commenter said that multiple quotes with the same timestamp could cause sequencing confusion and suggested that the Commission provide more specificity to address this concern.[1022]

The Commission believes that timestamps are of particular importance in a decentralized consolidation model because competing consolidators will be generating consolidated market data products individually. Market participants must be able to understand the market at the time their orders are represented and executed. Further, timestamps help to ensure that competing consolidators are accurately calculating and disseminating consolidated market data products. Therefore, the Commission is adopting these requirements as proposed.

The exclusive SIPs' timestamp information is similar to what is required of competing consolidators under Rules 614(d)(4)(i) and (iii). The timestamp requirement will allow subscribers to ascertain how quickly the competing consolidator can receive data from the exchanges, transmit that data between the exchange's data center and its aggregation center, and aggregate and disseminate its consolidated market data product to subscribers (its realized latency). The Commission also believes that this information will provide transparency that should help subscribers evaluate a potential competing consolidator or determine whether an existing competing consolidator continues to meet their needs.

1019 See Capital Group Letter at 4.

1020 See Data Boiler Letter I at 53.

1021 See TD Ameritrade Letter at 13.

1022 See id.

The Commission does not think that the addition of timestamps on competing consolidators' consolidated market data products will cause sequencing confusion. Rule 614(d)(4) requires each competing consolidator to affix its own timestamps to the information collected in Rule 614(d)(1): (i) upon receipt from each SRO at the exchange's or association's data center; (ii) upon receipt of such information at its aggregation mechanism; and (iii) upon dissemination of consolidated market data to customers. As noted above, currently, the exclusive SIPs similarly timestamp information. The timestamp requirement should not introduce any new sequencing confusion. Instead, this timestamping requirement should help subscribers understand a competing consolidator's performance in generating consolidated market data products. Competing consolidators will have different systems to collect, calculate, and disseminate the data they receive from the SROs and their timestamps will help market participants measure latencies.

(c) Rules 614(d)(5) and (6): Monthly Website Publication of Performance and Operational Information

(i) Proposal

Proposed Rule 614(d)(5) required each competing consolidator to publish prominently on its website, within 15 calendar days after the end of each month, certain performance metrics. All information posted pursuant to proposed Rule 614(d)(5) must be publicly posted in downloadable files and must remain free and accessible (without any encumbrances or restrictions) by the general public on the website for a period of not less than three years from the initial date of posting.

In particular, proposed Rule 614(d)(5) required the publication of the following performance metrics: (i) capacity statistics (such as system tested capacity, system output

capacity, total transaction capacity, and total transaction peak capacity); (ii) message rate and total statistics (such as peak output rates on the following bases: 1-millisecond, 10-millisecond, 100-millisecond, 500-millisecond, 1-second, and 5-second); (iii) system availability statistics (for example, whether system up-time has been 100% for the month and cumulative amount of outage time); (iv) network delay statistics (for example, today under a TCP-IP network, network delay statistics would include quote and trade zero window size events, quote and trade TCP retransmit events, and quote and trade message total); and (v) latency statistics (with distribution statistics up to the 99.99th percentile) for (1) when a national securities exchange or national securities association sends an inbound message to a competing consolidator network and when the competing consolidator network receives the inbound message; (2) when the competing consolidator network receives the inbound message and when the competing consolidator network sends the corresponding consolidated message to a subscriber; and (3) when a national securities exchange or national securities association sends an inbound message to a competing consolidator network and when the competing consolidator network sends the corresponding consolidated message to a subscriber.

Proposed Rule 614(d)(6) required each competing consolidator to publish prominently on its website, within 15 calendar days after the end of each month, information on: (i) data quality issues (such as delayed message publication, publication of duplicative messages, and message inaccuracies); (ii) system issues (such as processing, connectivity, and hardware problems); (iii) any clock synchronization protocol utilized; (iv) for the clocks used to generate the timestamps described in Rule 614(d)(4), clock drift averages and peaks and number of instances of clock drift greater than 100 microseconds; and (v) vendor alerts (such as holiday reminders and testing dates). All information posted pursuant to proposed Rule 614(d)(6) must be publicly posted and

must remain free and accessible (without any encumbrances or restrictions) by the general public on the website for a period of not less than three years from the initial date of posting.

(ii) Final Rule and Response to Comments

The Commission received several comments regarding proposed Rules 614(d)(5) and (d)(6).[1023] Five commenters supported the proposed Rules.[1024] One commenter objected to the proposed rules.[1025] One commenter requested guidance from the Commission relating to broker-dealers' use of the required information.[1026]

Several commenters supported requiring competing consolidators to disclose the information required by the proposed rules,[1027] and some commenters said this information would be useful in choosing a competing consolidator.[1028] One commenter said that such transparency could help keep costs down,[1029] and another commenter stated that the publication of performance metrics, in combination with competition, would "advance the objective of promoting useful and widely available market data for a range of market participants."[1030] One commenter indicated that the information and frequency with which it would be provided were

---

1023 See Clearpool Letter at 9; IntelligentCross Letter at 5; IEX Letter at 8; ACS Execution Services Letter at 6; Data Boiler Letter I at 53; STANY Letter II at 6; FINRA Letter at 5.

1024 See Clearpool Letter at 9; IntelligentCross Letter at 5; IEX Letter at 8; ACS Execution Services Letter at 6; Data Boiler Letter I at 53.

1025 See STANY Letter II at 6.

1026 See FINRA Letter at 5.

1027 See Clearpool Letter at 9; IntelligentCross Letter at 5; IEX Letter at 8.

1028 See IntelligentCross Letter at 5; ACS Execution Services Letter at 6.

1029 See ACS Execution Services Letter at 6.

1030 IEX Letter at 8.